Code No: 138GY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
SENSORS AND TRANSDUCERS
(Common to EEE, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used Papers 2024

PART – A

(25 Marks)

1.a) Classify different transducers in measurements. [2]
b) Differentiate between sensors and transducers. [3]
c) Illustrate how mathematical models can be used to analyze the dynamic characteristics of transducers. [2]
d) How does a second order transducer respond to a step input? [3]
e) Explain the concept of temperature compensation in strain gauge measurements. [2]
f) Discuss the advantages and limitations of using a potentiometer for measurement applications. [3]
g) State the principle of operation of mutual inductive transducers. [2]
h) What are the advantages of using an LVDT for precision measurements? [3]
i) How Hall Effect is applied in the operation of Hall Effect transducers. [2]
j) Discuss the key characteristics of nano sensors. [3]

PART – B

(50 Marks)

2.a) Explain the fundamental process involved in the basic method of measurement. Provide an example to illustrate this process.
b) Discuss the classification of sensors based on the measurement of physical quantities. Provide detailed explanations of specific sensors within each classification and their applications. [5+5]

OR

3.a) Define errors in measurements and discuss the difference between systematic and random errors. Explain how each type of error can impact the accuracy of measurements.
b) List and explain the basic requirements of a transducer in a measurement system. Why is sensitivity to input physical quantities crucial for transducers? [5+5]

4.a) Define static characteristics of transducers and discuss their importance in measurement Systems.
b) What is a mathematical model of a transducer, and how is it developed? Discuss the key parameters and variables involved in creating a mathematical model for transducers.[5+5]

OR

5.a) Discuss the frequency response of first-order transducers. Analyze how the frequency of the input signals affects the performance and sensitivity of first-order transducers in different applications.

b) Explain the response of a zero-order transducer to an impulse input. Discuss how the transducer reacts to sudden changes in input and its ability to capture instantaneous variations in the measured quantity. [5+5]

6.a) Describe the working principle of a potentiometer and its significance in precision measurements.

b) Explain the concept of operation of strain gauges and their role in measuring mechanical deformation. [5+5]

OR

7.a) Discuss in detail the construction and characteristics of resistance thermometers compared to thermistors.

b) How does a hot wire anemometer measure airflow velocity, and what are its advantages over other airflow measurement techniques? [5+5]

8.a) Explain the principle of operation of a self-inductive transducer and its applications in electrical measurement systems.

b) Describe the construction and working principle of an LVDT and explain how it converts mechanical displacement into electrical signals. [5+5]

OR

9.a) Describe the operating principles of accelerometers and their role in measuring acceleration in dynamic systems.

b) Explain how changes in the area of capacitive plates affect the capacitance and signal output. [5+5]

10.a) Discuss the design and operation of Hall Effect transducers in detecting current, position, and speed.

b) Explain the working principle of fiber optic sensors and how they utilize light propagation for sensing applications. [5+5]

OR

11.a) Describe the construction and properties of film sensors for measuring physical quantities.

b) Explain the concept of digital transducers and how they convert physical quantities into digital signals for processing and communication. [5+5]

---ooOoo---

Used Papers 2024